

Time Series Analysis

Chapter 14: Multivariate GARCH models



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CCC**: Constant Conditional Correlation (model)
- **CEE**: Central and Eastern Europe
- **CET**: Central European Time
- **COVID-19**: COronaVIrus Disease 2019
- **DCC**: Dynamic Conditional Correlation
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **FHS**: Filtered Historical Simulation
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **LR**: Likelihood Ratio (test)
- **MGARCH**: Multivariate GARCH
- **OLS**: Ordinary Least Squares
- **QML**: Quasi-Maximum Likelihood
- **SE**: Standard Error
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VAR**: Vector AutoRegression
- **VEC**: Vector (half-vectorised) GARCH model
- **VECM**: Vector Error Correction Model

Motivation for the chapter

- **Question**: how do the risks of several assets move **together**? Chapter 5 modelled the volatility of one series at a time
  - the risk of a portfolio, a hedge or the spread of a crisis depends on **covariances and correlations**, and these change in time
- **Route** of the chapter
  - the conditional covariance matrix $\mathbf{H}_t$ and its conditions
  - the VEC and BEKK models; the number of parameters
  - CCC and DCC: volatilities and correlations estimated in two steps
  - applications: correlations in crises, portfolio VaR 1%, hedge ratios
- **Self-study chapter**
  - read the slides in order, redo the worked examples, run the notebook, then answer the self-assessment and the quiz
  - prerequisites: Chapter 5 (GARCH), Chapter 6 (VAR models), Chapter 7 (cointegration)

Learning outcomes

- Explain why portfolio risk, hedging and contagion need a model for time-varying covariances
- Write the VEC, BEKK, CCC and DCC models and count their parameters
- Compute one step of the DCC recursion by hand and interpret a, b and $a + b$
- Estimate a DCC model in two steps and test it against constant correlations
- Compute and backtest a portfolio VaR 1% from $\mathbf{H}_t$
- Compute dynamic minimum-variance hedge ratios and their hedging effectiveness

Reading and tools

- Textbook background: Huang and Petukhina (2022) (GARCH models); surveys: Bauwens, Laurent and Rombouts (2006), Silvennoinen and Teräsvirta (2009)
  - the original papers: Bollerslev, Engle and Wooldridge (1988) (VEC), Engle and Kroner (1995) (BEKK), Bollerslev (1990) (CCC), Engle (2002) (DCC)
- Python Quantlets of this chapter: Quantlets/Ch_14
  - step 1 (univariate GARCH) uses the `arch` package; step 2 (DCC) is written out in about 40 lines of `numpy`, because `arch` has no DCC
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

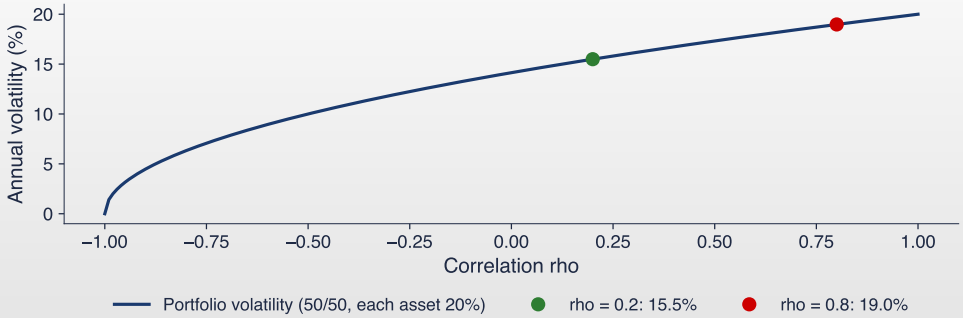
Three applications that require covariances

- **Portfolio risk**
  - the variance of a portfolio with weights $\mathbf{w}$ is $\sigma_p^2 = \mathbf{w}^\top \mathbf{H}\mathbf{w}$ Markowitz (1952)
  - $\mathbf{w}$: the vector of portfolio weights; $\mathbf{H}$: the covariance matrix of the returns; σ_p^2: the portfolio variance
  - two assets: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\,\sigma_1\sigma_2$, with volatilities σ_1, σ_2 and correlation ρ
  - the correlation ρ decides how much diversification helps
- **Hedging**
  - how many units of an asset to sell to offset the risk of another
  - the answer is a ratio of a covariance to a variance (Section 7)
- **Contagion**
  - do markets move more closely together in a crisis?
  - if correlations rise exactly when volatility rises, diversification fails when it is needed most Longin and Solnik (2001)
- A univariate GARCH (Chapter 5) gives $\sigma_{1,t}$ and $\sigma_{2,t}$; this chapter adds the time-varying ρ_t

Worked example: correlation and portfolio risk

- Two assets with annual volatility 20% each, weights 50% and 50%
- $\rho = 0.2$: $\sigma_p^2 = 0.25 \cdot 400 + 0.25 \cdot 400 + 2 \cdot 0.25 \cdot 0.2 \cdot 400 = 240$, so $\sigma_p = 15.5\%$
- $\rho = 0.8$: $\sigma_p^2 = 200 + 2 \cdot 0.25 \cdot 0.8 \cdot 400 = 360$, so $\sigma_p = 19.0\%$
- The same assets and weights: the risk of the portfolio rises by more than a fifth only because the correlation rose
  - a risk model that keeps ρ fixed at its calm-period value underestimates the risk in a crisis

Portfolio volatility against the correlation



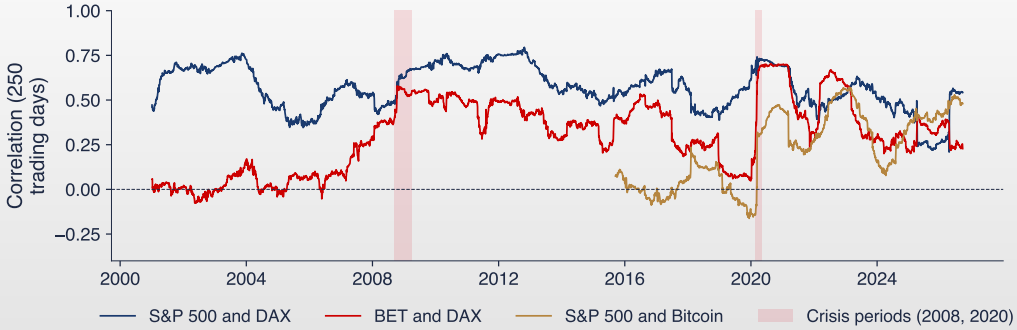
- A 50/50 portfolio of two assets with 20% volatility each; the points are the two cases of the worked example

TSA_ch14_correlations

Interpreting the diversification curve

- $\rho = 1$: no diversification, $\sigma_p = 20\%$; $\rho = 0$: $\sigma_p = 20/\sqrt{2} \approx 14.1\%$; $\rho = -1$: the risk disappears
- The curve is concave: the same rise of ρ adds more volatility at low correlations than at high ones
- The question of the chapter: does ρ stay put in real markets, or does it move with the state of the market?

Time-varying correlations



- Correlation of daily log returns over rolling windows of 250 trading days (about one year), dated at the window end
  - returns on the days on which both markets trade; shaded: September 2008–March 2009 and February–April 2020

TSA_ch14_correlations

Interpreting the rolling correlations

- S&P 500 and DAX: between 0.21 (April 2026) and 0.80 (October 2012); full-sample value 0.60
  - the euro-area debt crisis of 2010–2012 tied the two markets even more closely than 2008
- BET and DAX: close to zero before 2007 (minimum -0.08), up to 0.70 in August 2020: the Romanian market became integrated with Europe after EU accession
- S&P 500 and Bitcoin: from -0.16 (December 2019) to 0.57 (February 2023); Bitcoin became a risk asset correlated with equities after 2020
- Correlations wander far from any constant value: a model must let them move

Asynchronous trading and the bias of daily correlations

- Markets close at different hours: Bucharest at 16:45 CET (17:45 local time), Frankfurt at 17:30 CET, New York at 22:00 CET; Bitcoin trades all day, every day
  - news that arrives after the European close shows up in the European prices only the next day
- Example: 9 April 2025, the pause of the US tariffs announced in the afternoon in New York
  - 9 April: S&P 500 $+9.1\%$, DAX -3.1%; 10 April: S&P 500 -3.5%, DAX $+4.4\%$
  - the same news, on two different days: the daily correlation is biased towards zero
- Correlations of daily and weekly returns since 2015: S&P 500 and BET 0.32 against 0.46; S&P 500 and DAX 0.56 against 0.68; BET and DAX 0.42 against 0.47
  - the gap is small for BET and DAX, which close almost together; remedies: weekly returns or returns between synchronous times

Recap: The role of multivariate volatility

- Portfolio variance $\mathbf{w}^\top \mathbf{H}\mathbf{w}$, hedge ratios and contagion all depend on covariances
- Rolling correlations of real markets move a lot: integration, crises, new asset classes
- Check the trading hours: asynchronous closes bias daily correlations towards zero

The model for N return series

- $\mathbf{r}_t = \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t = \mathbf{H}_t^{1/2}\boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t$ i.i.d. with mean $\mathbf{0}$ and covariance $\mathbf{I}_N$
  - $\mathbf{r}_t$: the vector of N returns on day t; $\boldsymbol{\mu}$: the means (or a VAR model for the mean, Chapter 6)
  - $\mathbf{H}_t^{1/2}$: a square root of $\mathbf{H}_t$, e.g. the Cholesky factor $\mathbf{L}_t$ with $\mathbf{L}_t\mathbf{L}_t^\top = \mathbf{H}_t$
- $\mathbf{H}_t = \mathrm{Var}(\mathbf{r}_t \mid \mathcal{F}_{t-1})$: the **conditional covariance matrix**, given the information $\mathcal{F}_{t-1}$ up to day $t-1$
  - diagonal: the conditional variances $h_{ii,t} = \sigma_{i,t}^2$ (Chapter 5)
  - off-diagonal: the conditional covariances $h_{ij,t}$; the conditional correlation is $\rho_{ij,t} = h_{ij,t}/\sqrt{h_{ii,t}h_{jj,t}}$
- A **multivariate GARCH** (MGARCH) model is a rule that computes $\mathbf{H}_t$ from past shocks $\boldsymbol{\varepsilon}_{t-1}$ and past $\mathbf{H}_{t-1}$

Two conditions on $\mathbf{H}_t$

- **Symmetry**
  - $h_{ij,t} = h_{ji,t}$, so only $N(N+1)/2$ elements are distinct
  - the operator vech stacks the lower triangle: $\text{vech}\begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix} = (h_{11}, h_{12}, h_{22})^\top$
  - $N = 2$: 3 elements; $N = 10$: 55; $N = 50$: 1 275
- **Positive definiteness**
  - $\mathbf{w}^\top \mathbf{H}_t \mathbf{w} > 0$ for every $\mathbf{w} \neq \mathbf{0}$, i.e. every portfolio has a positive variance
  - $N = 2$: $h_{11,t} > 0$ and $h_{11,t} h_{22,t} - h_{12,t}^2 > 0$, i.e. $|\rho_{12,t}| < 1$
  - example: $h_{11} = 4$, $h_{22} = 1$, $h_{12} = 2.5$ gives $4 - 6.25 < 0$: an impossible "correlation" of 1.25
- A good MGARCH model guarantees both conditions for **every** t, with few parameters

Estimation by maximum likelihood

- If $\boldsymbol{\eta}_t$ follows the multivariate Normal distribution, the log-likelihood is
  - $\ell(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(N\log 2\pi + \log|\mathbf{H}_t| + \boldsymbol{\varepsilon}_t^\top \mathbf{H}_t^{-1}\boldsymbol{\varepsilon}_t\right)$
  - $|\mathbf{H}_t|$ is the determinant; for $N = 1$ this is the GARCH likelihood of Chapter 5
- Returns have fat tails, so the Normal likelihood is used as **quasi-maximum likelihood** (QML)
  - the estimates stay consistent; the standard errors need a robust (sandwich) correction
- Every evaluation of $\ell(\theta)$ needs $\mathbf{H}_t^{-1}$ and $|\mathbf{H}_t|$ for all t: the cost grows fast with N and with the number of parameters

Recap: The multivariate GARCH framework

- $\mathbf{H}_t$: conditional variances on the diagonal, covariances off the diagonal; $\rho_{ij,t} = h_{ij,t}/\sqrt{h_{ii,t}h_{jj,t}}$
- Conditions: symmetry ($N(N+1)/2$ distinct elements) and positive definiteness
- Estimation by (quasi-)maximum likelihood with the multivariate Normal density

The VEC model

- Bollerslev, Engle and Wooldridge (1988): every element of $\mathbf{H}_t$ depends on every past squared shock, cross-product and element of $\mathbf{H}_{t-\mathbf{1}}$
  - VEC(1,1): $\text{vech}(\mathbf{H}_t) = \mathbf{c} + \mathbf{A}\,\text{vech}(\boldsymbol{\varepsilon}_{t-\mathbf{1}}\boldsymbol{\varepsilon}_{t-\mathbf{1}}^{\top}) + \mathbf{B}\,\text{vech}(\mathbf{H}_{t-\mathbf{1}})$
  - $\boldsymbol{\varepsilon}_{t-\mathbf{1}}\boldsymbol{\varepsilon}_{t-\mathbf{1}}^{\top}$: the matrix of yesterday's squared shocks and cross-products; $\mathbf{c}$: a vector of constants; $\mathbf{A}$: the reaction to shocks; $\mathbf{B}$: the persistence
  - with $k = N(N+1)/2$: $\mathbf{c}$ has k elements, $\mathbf{A}$ and $\mathbf{B}$ are $k \times k$: in total $k + 2k^{\mathbf{2}}$ parameters
- $N = 2$: $k = 3$ and 21 parameters; $N = 10$: 6 105
  - and positive definiteness of $\mathbf{H}_t$ is **not** guaranteed: it needs complicated restrictions
- **Diagonal VEC**
  - $h_{ij,t} = c_{ij} + a_{ij}\varepsilon_{i,t-\mathbf{1}}\varepsilon_{j,t-\mathbf{1}} + b_{ij}h_{ij,t-\mathbf{1}}$
  - each element is a GARCH(1,1) of its own; $3k$ parameters (9 for $N = 2$); still no guarantee of positive definiteness

The BEKK model

- Engle and Kroner (1995): $\mathbf{H}_t = \mathbf{C}\mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-\mathbf{1}} \boldsymbol{\varepsilon}_{t-\mathbf{1}}^\top \mathbf{A} + \mathbf{B}^\top \mathbf{H}_{t-\mathbf{1}} \mathbf{B}$
  - BEKK: Baba, Engle, Kraft and Kroner; $\mathbf{C}$ lower triangular, $\mathbf{A}$ and $\mathbf{B}$ are $N \times N$
  - $\mathbf{A}$: the reaction to yesterday's shocks; $\mathbf{B}$: the persistence of $\mathbf{H}_{t-\mathbf{1}}$; c_{ij}^*: the elements of $\mathbf{C}\mathbf{C}^\top$
- **Positive definite by construction**
  - each term is a "square" $\mathbf{X}^\top \mathbf{M} \mathbf{X}$ of a positive (semi)definite matrix $\mathbf{M}$
  - $\mathbf{M}$: $\mathbf{I}$, $\boldsymbol{\varepsilon}_{t-\mathbf{1}} \boldsymbol{\varepsilon}_{t-\mathbf{1}}^\top$ or $\mathbf{H}_{t-\mathbf{1}}$; $\mathbf{X}$: $\mathbf{C}^\top$, $\mathbf{A}$ or $\mathbf{B}$
  - parameters: $N(N+1)/2 + 2N^{\mathbf{2}}$, i.e. 11 for $N = 2$ and 255 for $N = 10$
- **Diagonal BEKK** ($\mathbf{A}$, $\mathbf{B}$ diagonal)
  - $h_{\mathbf{12},t} = c_{\mathbf{12}}^* + a_{\mathbf{11}} a_{\mathbf{22}}\, \varepsilon_{\mathbf{1},t-\mathbf{1}} \varepsilon_{\mathbf{2},t-\mathbf{1}} + b_{\mathbf{11}} b_{\mathbf{22}}\, h_{\mathbf{12},t-\mathbf{1}}$
  - **scalar BEKK**: $\mathbf{A} = a\mathbf{I}$, $\mathbf{B} = b\mathbf{I}$, the same dynamics for all elements
- The off-diagonal elements of a full $\mathbf{A}$ measure **volatility spillovers**: a shock to asset 1 raises the variance of asset 2

Worked example: one step of a diagonal BEKK

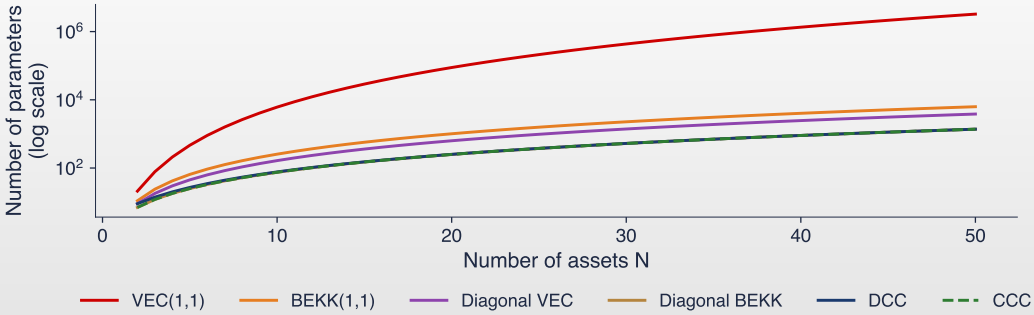
- Given: $c_{12}^{*} = 0.02$, $a_{11} = 0.30$, $a_{22} = 0.25$, $b_{11} = 0.94$, $b_{22} = 0.95$, $h_{12,t-1} = 0.5$; yesterday's shocks $\varepsilon_{1,t-1} = -2$, $\varepsilon_{2,t-1} = -1.5$
- Step 1: the coefficients $a_{11}a_{22} = 0.075$ and $b_{11}b_{22} = 0.893$
- Step 2: the cross-product of shocks $\varepsilon_{1,t-1}\varepsilon_{2,t-1} = 3.000$ (both negative: a joint fall)
- Step 3: $h_{12,t} = 0.020 + 0.075 \cdot 3.000 + 0.893 \cdot 0.5 = 0.020 + 0.225 + 0.4465 = 0.6915$
- A joint fall raises the covariance from 0.5 to 0.6915; a shock of opposite signs would have lowered it
  - note: a diagonal BEKK has no spillovers, because $a_{12} = a_{21} = 0$

The curse of dimensionality

Model	$N = 2$	$N = 5$	$N = 10$	$N = 50$
VEC(1,1)	21	465	6 105	3 252 525
Diagonal VEC	9	45	165	3 825
BEKK(1,1)	11	65	255	6 275
Diagonal BEKK	7	25	75	1 375
Scalar BEKK	5	17	57	1 277
CCC	7	25	75	1 375
DCC	9	27	77	1 377

- Number of parameters of the variance equation (the means μ are not counted); CCC and DCC: 3 GARCH parameters per asset plus the $N(N-1)/2$ correlations, and 2 more for DCC
- A full VEC for 50 assets has more parameters than any sample has observations

Parameters against the number of assets



Number of parameters of the variance equation for $N = 2, \ldots, 50$ assets, log scale

TSA_ch14_parameters

Interpreting the parameter counts

- VEC grows like $N^4/2$, full BEKK like $2N^2$: both become impossible to estimate beyond a handful of assets
- CCC and DCC grow like $N^2/2$, but the correlations are **not** found by numerical optimisation: they are sample correlations (Section 4)
  - the DCC optimiser works on only 2 parameters, a and b, whatever N

Recap: VEC and BEKK

- VEC: the most general linear model; $k + 2k^2$ parameters, no positive-definiteness guarantee
- BEKK: positive definite by construction; off-diagonal a_{ij} measure volatility spillovers
- Diagonal and scalar versions cut the parameters but remove the spillovers

Splitting volatilities and correlations

- Decomposition: $\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t, \quad \mathbf{D}_t = \mathrm{diag}(\sigma_{1,t}, \ldots, \sigma_{N,t})$
  - $\mathbf{R}_t$: the conditional correlation matrix (ones on the diagonal); element by element: $h_{ij,t} = \rho_{ij,t}\, \sigma_{i,t}\sigma_{j,t}$
  - each $\sigma_{i,t}$ is a univariate GARCH(1,1) Bollerslev (1986): $\sigma^2_{i,t} = \omega_i + \alpha_i \varepsilon^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}$
- **Standardised residuals**
  - $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$
  - their conditional covariance matrix is exactly $\mathbf{R}_t$
- **CCC** (constant conditional correlation, Bollerslev (1990))
  - $\mathbf{R}_t = \mathbf{R}$ for all t
  - $\mathbf{H}_t$ is positive definite whenever $\mathbf{R}$ is; $\mathbf{R}$ is estimated by the sample correlation of the $z_{i,t}$
  - covariances still move, but only through the volatilities; the rolling correlations of Section 1 reject this

Robert Engle and the DCC model



Robert F. Engle (b. 1942), Nobel Prize 2003

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons

- Engle (2002): let the correlations follow a GARCH-like recursion
  - $\mathbf{Q}_t = (1 - a - b)\,\bar{\mathbf{Q}} + a\,\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\top + b\,\mathbf{Q}_{t-1}$
  - $\mathbf{Q}_t$: an auxiliary matrix with elements $q_{ij,t}$; $\mathbf{z}_{t-1}$: yesterday's vector of standardised residuals
  - $\rho_{ij,t} = q_{ij,t} / \sqrt{q_{ii,t}\, q_{jj,t}}$: the rescaling puts ones on the diagonal
- $\bar{\mathbf{Q}}$: the unconditional correlation of $\mathbf{z}_t$, set to its sample value (**correlation targeting**)
  - $a \geq 0$: reaction to yesterday's joint shock; $b \geq 0$: persistence; $a + b < 1$: mean reversion to $\bar{\mathbf{Q}}$
- $a = b = 0$ gives back CCC

Two-step estimation

- The log-likelihood splits into a volatility part and a correlation part: $\ell = \ell_V(\theta_{\mathbf{1}}) + \ell_C(\theta_{\mathbf{1}}, a, b)$
  - $\theta_{\mathbf{1}}$: the GARCH parameters of all series; ℓ_V depends only on them, ℓ_C also on a and b
- **Step 1**
  - fit a GARCH(1,1) to each series separately (Chapter 5)
  - keep $\hat{\sigma}_{i,t}$ and $\hat{z}_{i,t} = \hat{\varepsilon}_{i,t}/\hat{\sigma}_{i,t}$
- **Step 2**
  - set $\bar{\mathbf{Q}}$ to the sample correlation of $\hat{\mathbf{z}}_t$ and maximise over (a, b) only
  - $\ell_C(a, b) = -\frac{\mathbf{1}}{\mathbf{2}} \sum_t \left(\log|\mathbf{R}_t| + \hat{\mathbf{z}}_t^\top \mathbf{R}_t^{-\mathbf{1}} \hat{\mathbf{z}}_t - \hat{\mathbf{z}}_t^\top \hat{\mathbf{z}}_t \right)$
- Consistent, fast, works for large N; the step-2 standard errors ignore the estimation error of step 1 (they are too small)
- **Test against CCC**
  - likelihood ratio $LR = 2(\ell_C^{DCC} - \ell_C^{CCC})$, compared with $\chi^{\mathbf{2}}(2)$ (5% critical value 5.99)
  - also Tse (2000)
  - ℓ_C^{DCC}, ℓ_C^{CCC}: the maximised correlation log-likelihoods of the two models; a large LR rejects constant correlation
  - $a = b = 0$ lies on the boundary of the parameter space, so the $\chi^{\mathbf{2}}(2)$ critical value is conservative

Worked example: one step of the DCC recursion

- Given: $a = 0.05$, $b = 0.93$, $\bar{q}_{\mathbf{12}} = 0.5$ (and $\bar{q}_{\mathbf{11}} = \bar{q}_{\mathbf{22}} = 1$), $\mathbf{Q}_{t-\mathbf{1}} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, yesterday $z_{\mathbf{1},t-\mathbf{1}} = -2$, $z_{\mathbf{2},t-\mathbf{1}} = -2.5$
- Step 1: $1 - a - b = 0.020$
- Step 2: $q_{\mathbf{12},t} = 0.020 \cdot 0.5 + 0.05 \cdot (-2)(-2.5) + 0.93 \cdot 0.5 = 0.010 + 0.250 + 0.465 = 0.725$
- Step 3: $q_{\mathbf{11},t} = 0.020 + 0.05 \cdot 4 + 0.93 = 1.150$; $\quad q_{\mathbf{22},t} = 0.020 + 0.05 \cdot 6.25 + 0.93 = 1.2625$
- Step 4: $\rho_{\mathbf{12},t} = 0.725/\sqrt{1.150 \cdot 1.2625} = 0.602$
  - a large joint fall raises tomorrow's correlation from 0.50 to 0.602; without step 3 the "correlation" $q_{\mathbf{12},t}$ would be 0.725

Interpreting a and b

- a is the **news** coefficient
  - how much one day's joint shock moves the correlation
  - typical daily values 0.01–0.05
- $a + b$ is the **persistence**
  - a deviation of $q_{ij,t}$ from $\bar{q}_{ij}$ shrinks by the factor $a + b$ each day
  - half-life: $\ln 0.5 / \ln(a + b)$ days; $a + b = 0.98$ gives 34 days, 0.99 gives 69 days
- The same reading as α and $\alpha + \beta$ in a GARCH(1,1) (Chapter 5)
- Variants: asymmetric DCC (joint falls move correlations more than joint rises) Cappiello, Engle and Sheppard (2006); corrected DCC Aielli (2013)

Recap: CCC and DCC

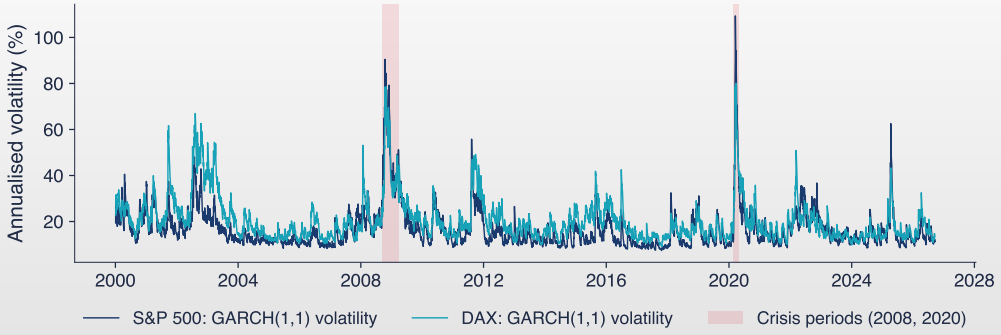
- $\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$: GARCH volatilities times a correlation matrix
- CCC: $\mathbf{R}$ constant; DCC: $\mathbf{Q}_t$ follows a GARCH-like recursion with parameters a, b
- Two steps: univariate GARCH, then (a, b) with correlation targeting; LR test against CCC

Step 1: two univariate GARCH models

Index	μ	ω	α	β	$\alpha+\beta$	kurt. r_t	kurt. z_t
S&P 500	0.063	0.027	0.122	0.857	0.980	10.5	1.8
DAX	0.067	0.030	0.098	0.886	0.984	6.2	1.4

- Daily log returns in %, 6 605 common trading days, 4 January 2000–18 September 2026 (EODHD); GARCH(1,1) with a constant mean, Gaussian QML; kurt.: excess kurtosis
- Standardising by $\hat{\sigma}_{i,t}$ removes most of the excess kurtosis, but not all: the tails of z_t are still fatter than Normal
- Correlation of the returns 0.599; correlation of the residuals $\bar{q}_{\mathbf{12}} = 0.575$, the CCC estimate

Step 1: conditional volatilities



⊡ One-step-ahead GARCH(1,1) volatility $\hat{\sigma}_{i,t}\sqrt{252}$, annualised, in %; shaded: the crises of 2008 and 2020

TSA_ch14_dcc_estimation

Interpreting the volatilities

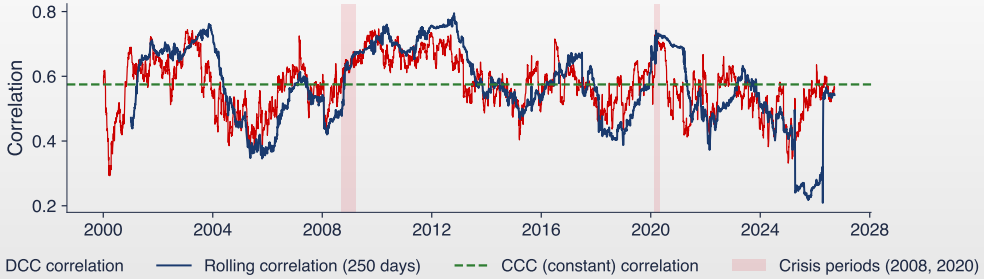
- Both volatilities peak in the same weeks: S&P 500 109% on 17 March 2020, DAX 80% on 25 March 2020
  - volatility clustering happens **jointly**: the covariance rises even with a constant correlation
- Persistence 0.980 and 0.984: volatility shocks fade slowly, as in Chapter 5
- Step 2 asks a different question: once the volatilities are removed, is the remaining correlation constant?

Step 2: the estimated DCC

Quantity	Estimate	SE
a	0.0152	0.0035
b	0.9766	0.0065
$a + b$	0.9918	
Half-life of a correlation shock (days)	85	
LR against CCC	131.8	
$\chi^2(2)$ critical value, 5%	5.99	

- a is small but well determined; $a + b$ is close to 1: correlation shocks last for months
- LR $= 131.8$ is far above 5.99: constant correlation is rejected
- SE: standard errors from the step-2 Hessian (too small, they ignore step 1)

DCC correlation of the S&P 500 and the DAX



- One-step-ahead DCC correlation $\hat{\rho}_{\mathbf{12},t}$, the 250-day rolling correlation and the CCC value $\bar{q}_{\mathbf{12}}$; daily, 4 January 2000–18 September 2026

TSA_ch14_dcc_estimation

Interpreting the DCC correlation

- The DCC correlation ranges from 0.29 (March 2000) to 0.75 (January 2010), around a mean of 0.57
- It follows the rolling correlation, but reacts within days instead of a year and has no window to choose
  - the rolling estimate is more erratic (standard deviation 0.122 against 0.083) and jumps when an extreme day enters or leaves the window
- The rolling estimate falls sharply in 2025–2026: the asynchronous days of April 2025 (Section 1) pull it down for exactly 250 days

Correlations in crises



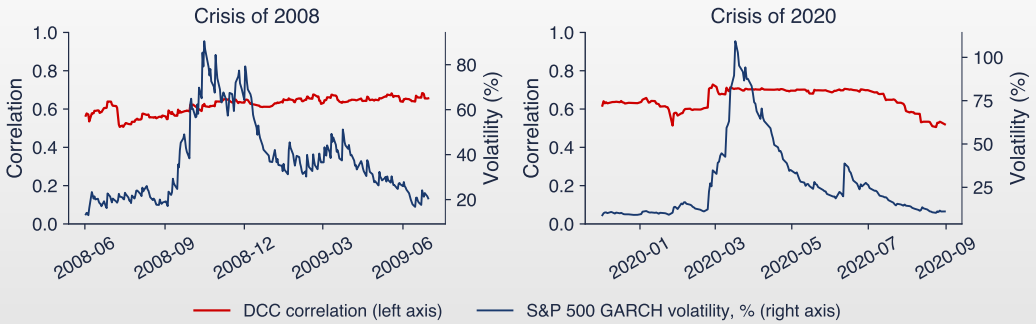
Lehman Brothers, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons

Period	$\bar{\rho}^{DCC}$	$\bar{\sigma}^{S\&P}$ (%)
Calm, 2017–2019	0.56	13
Sept 2008–March 2009	0.63	51
Feb–April 2020	0.70	58

- Average DCC correlation and average annualised S&P 500 volatility in each period
- Correlations are higher in crises, but volatility rises far more
- Caution Forbes and Rigobon (2002): the sample correlation rises mechanically when volatility rises, even if the link is unchanged; DCC reduces this effect because it works with z_t, not with r_t

DCC correlation in the crises of 2008 and 2020



⊡ DCC correlation of the S&P 500 and the DAX (left axis) and the GARCH volatility of the S&P 500 (right axis)

TSA_ch14_dcc_estimation

Interpreting the two crises

- 2008: volatility peaks at 90% on 16 October 2008, while the correlation drifts up from 0.57 to 0.68 only by 23 June 2009
- 2020: the correlation jumps to 0.73 on 28 February 2020, before the volatility peak (109% on 17 March 2020)
  - a global shock hit both markets on the same days: joint falls drive the DCC recursion
- Correlations and volatilities need not peak together: modelling them separately, as DCC does, shows the difference

Recap: S&P 500 and DAX

- Step 1: persistent GARCH volatilities that peak together
- Step 2: $a + b = 0.9918$; LR 131.8: the correlation is not constant
- Higher correlations in 2008 and 2020, but the largest change is in volatility

A five-asset DCC

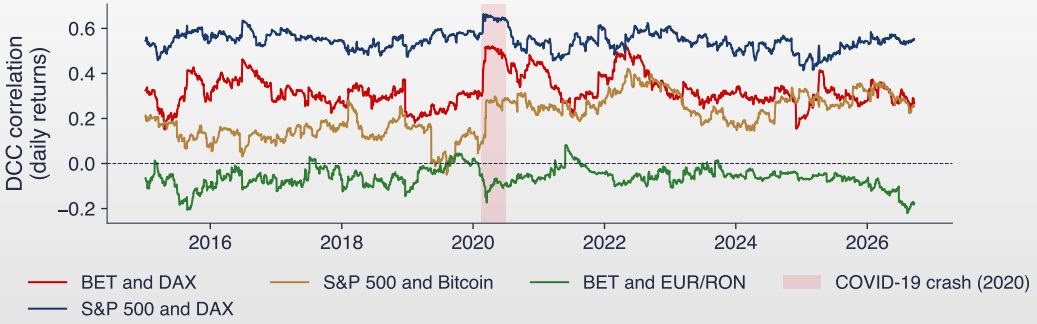


The Bucharest Stock Exchange

Photo: Corina Chitu (2024); CC BY-SA 4.0; Wikimedia Commons

- Daily log returns on the 2 815 days on which all five markets trade, from 6 January 2015
  - EUR/RON: the BNR reference rate (fixed at 13:00 Bucharest time); BET: Bucharest Stock Exchange; Bitcoin: close of the day (UTC)
- Step 1: GARCH persistence S&P 500 0.961, DAX 0.965, BET 0.941, EUR/RON 0.985, Bitcoin 0.958
- Step 2: $\hat{a} = 0.0088$ (SE 0.0019), $\hat{b} = 0.9782$ (SE 0.0062); half-life 53 days; LR against CCC 88.5
  - 10 correlation pairs, one pair of dynamics parameters (a, b) for all of them

Dynamic correlations of four pairs



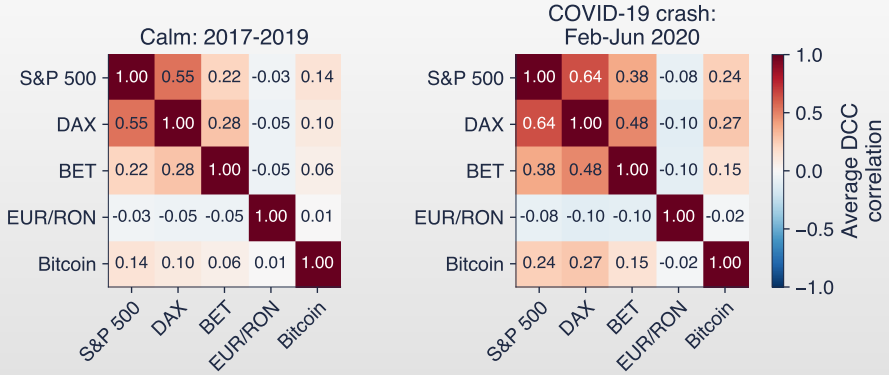
One-step-ahead DCC correlations from the five-asset model; shaded: the COVID-19 crash, mid-February–June 2020

TSA_ch14_dcc_panel

Interpreting the four pairs

- S&P 500 and DAX: the strongest link, mean 0.54; BET and DAX: mean 0.32, between 0.15 and 0.53 (April 2022)
- S&P 500 and Bitcoin: from -0.05 to 0.42 (May 2022); mean 0.21
  - the diversification benefit of Bitcoin shrank after 2020
- BET and EUR/RON: close to zero (mean -0.06); a weaker leu tends to come with a weaker BET, a small negative link
- One (a, b) for all pairs: pairs with different speeds are forced to the same rhythm, the price of DCC's parsimony

Average correlations: calm against crisis



- Average one-step-ahead DCC correlation matrix: 2017–2019 and mid-February–June 2020 (89 common trading days)

TSA_ch14_dcc_panel

Interpreting the two matrices

- The average pairwise correlation rises from 0.12 to 0.19
  - BET and DAX: 0.28 $\rightarrow$ 0.48; S&P 500 and BET: 0.22 $\rightarrow$ 0.38; DAX and Bitcoin: 0.10 $\rightarrow$ 0.27
- The largest rises are for the BET and for Bitcoin, which are weakly linked to the large markets in calm times: exactly the assets that were supposed to diversify
- EUR/RON stays near zero or slightly negative with all equity markets: the only diversifier in this set

Recap: Five markets

- One DCC for 5 assets: 2 dynamics parameters for 10 correlations
- Correlations of the Romanian market with Europe and of Bitcoin with equities move a lot
- In the 2020 crash the weak links strengthened most: diversification weakens in crises

From $\mathbf{H}_t$ to the VaR of a portfolio

- Portfolio return $r_{p,t} = \mathbf{w}^\top \mathbf{r}_t$; conditional mean $\mu_p = \mathbf{w}^\top \boldsymbol{\mu}$ and volatility $\sigma_{p,t} = \sqrt{\mathbf{w}^\top \mathbf{H}_t \mathbf{w}}$
- **VaR 1%**
  - the loss exceeded with probability 1%, $\text{VaR}_{0.01,t} = -q_{0.01}(r_{p,t}) = -(\mu_p + q\,\sigma_{p,t})$
  - Normal: $q = z_{0.01} = -2.326$
  - **filtered historical simulation** (FHS): q = the empirical 1% quantile of the past standardised portfolio returns $(r_{p,t} - \mu_p)/\sigma_{p,t}$
- Three choices of $\mathbf{H}_t$: DCC; CCC (GARCH volatilities with a constant correlation); **static** (one sample covariance matrix for all days)

Worked example: VaR 1% of two assets

- Today's forecasts: $\sigma_1 = 1.5\%$, $\sigma_2 = 1.2\%$ (daily), weights 50% and 50%, mean 0
- Calm, $\rho_t = 0.3$: $\sigma_p^2 = 0.25 \cdot 2.25 + 0.25 \cdot 1.44 + 2 \cdot 0.25 \cdot 0.3 \cdot 1.8$, so $\sigma_p = 1.092\%$ and VaR 1% $= 2.326 \cdot 1.092 = 2.54\%$
- Crisis, $\rho_t = 0.9$ (the same volatilities): $\sigma_p = 1.316\%$ and VaR 1% $= 3.06\%$
  - the jump of the correlation alone adds a fifth to the VaR; in a real crisis the volatilities rise too
- For a portfolio of 1 million lei: VaR 1% of about 25.4 thousand lei in calm times and 30.6 thousand lei in the crisis case

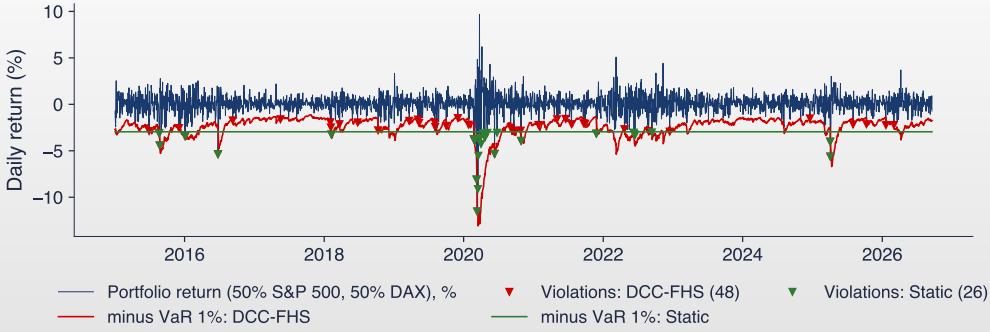
Backtesting a VaR (1/2): frequency

- A **violation** on day t
  - $r_{p,t} < -\text{VaR}_{\mathbf{0.01},t}$
  - a correct VaR 1% has violations on 1% of the days, at random times
- **Kupiec test** Kupiec (1995) (frequency)
  - x violations in n days, $\hat{p} = x/n$
  - $LR_{uc} = -2 \log \dfrac{(1-0.01)^{n-x}\, 0.01^{x}}{(1-\hat{p})^{n-x}\, \hat{p}^{x}} \sim \chi^{\mathbf{2}}(1)$; reject at 5% if $LR_{uc} > 3.84$
  - uc: unconditional coverage; the ratio compares the likelihood of the violations under the rate 1% with that under the observed rate $\hat{p}$

Backtesting a VaR (2/2): independence

- **Christoffersen test** Christoffersen (1998) (independence)
  - is a violation more likely the day after a violation?
  - compares $\pi_{11} = P(\text{violation} \mid \text{violation yesterday})$ with $\pi_{01} = P(\text{violation} \mid \text{no violation yesterday})$
- The statistic
  - $LR_{ind} = -2\log \dfrac{(1-\hat{\pi})^{n_{00}+n_{10}}\,\hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\,\hat{\pi}_{01}^{n_{01}}\,(1-\hat{\pi}_{11})^{n_{10}}\,\hat{\pi}_{11}^{n_{11}}} \sim \chi^2(1)$
  - n_{ij}: the number of days in state j after a day in state i (1: violation, 0: none); $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$
  - $\hat{\pi}$: the overall violation rate; the numerator assumes $\pi_{01} = \pi_{11}$ (no clustering); a small p-value rejects independence
- **Out-of-sample design**
  - estimate all parameters on 2000–2014 (3 712 days), then run the filters forward on 2015–2026 (2 893 days) without re-estimating

VaR 1% of an S&P 500 and DAX portfolio, 2015–2026



⊡ Daily return of a 50/50 portfolio and minus VaR 1% from DCC with filtered historical simulation and from a static covariance matrix; triangles: violations

TSA_ch14_portfolio_var

Interpreting the backtest

Model	Violations	Rate	LR_{uc}	p	p indep.	Feb–Apr 2020
DCC, Normal	65	2.25%	33.6	< 0.001	0.003	4
DCC, FHS	48	1.66%	10.6	0.001	0.008	4
CCC, Normal	57	1.97%	21.4	< 0.001	0.030	4
Static, Normal	26	0.90%	0.3	0.577	0.001	10

- Expected: about 29 violations in 2 893 days
- Dynamic models with Normal quantiles have too many violations: the residuals have fat tails (Chapter 5); FHS ($q = -2.59$) brings the rate down to 1.66%
- The static VaR passes the frequency test, but its violations cluster: 10 in February–April 2020 alone, and after a violation the next day is a violation with probability 11.5%
- The static VaR is too high in calm years (2.95% every day) and far too low in a crash; a dynamic VaR adapts in both directions

Recap: Portfolio VaR

- $\sigma_{p,t} = \sqrt{\mathbf{w}^\top \mathbf{H}_t \mathbf{w}}$; VaR 1% $= -(\mu_p + q\,\sigma_{p,t})$
- Backtest out of sample: Kupiec (frequency) and Christoffersen (independence)
- Dynamic covariances fix the clustering of violations; fat tails need a better quantile than the Normal

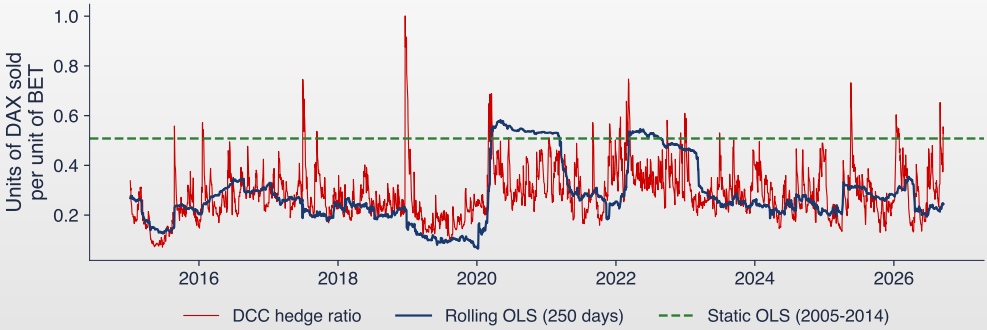
The minimum-variance hedge ratio

- Hold one unit of asset s (e.g. Romanian shares, the BET) and sell h units of a hedging asset f (e.g. a DAX futures contract)
  - hedged return $r_{s,t} - h\, r_{f,t}$, with variance $\sigma_s^2 - 2h\,\sigma_{sf} + h^2\sigma_f^2$
  - σ_s^2, σ_f^2: the variances of the two returns; σ_{sf}: their covariance
- Setting the derivative in h to zero: $h^* = \dfrac{\sigma_{sf}}{\sigma_f^2} = \rho\,\dfrac{\sigma_s}{\sigma_f}$ Ederington (1979)
  - static h^*: the OLS slope of r_s on r_f
  - **dynamic**: $h_t^* = h_{sf,t}/h_{ff,t}$ from an MGARCH model Kroner and Sultan (1993); $h_{sf,t}$, $h_{ff,t}$: elements of $\mathbf{H}_t$
- **Hedging effectiveness**
  - $HE = 1 - \mathrm{Var}(r_s - h\, r_f)/\mathrm{Var}(r_s)$, the share of the variance removed
  - with $h = h^*$ constant: $HE = \rho^2$; a hedge is only as good as the correlation

Worked example: a hedge ratio

- Given: $\sigma_s = 1.2\%$, $\sigma_f = 1.0\%$ (daily), $\rho = 0.8$
- Step 1: $\sigma_{sf} = \rho\,\sigma_s\sigma_f = 0.8 \cdot 1.2 \cdot 1.0 = 0.96$
- Step 2: $h^* = \sigma_{sf}/\sigma_f^2 = 0.96/1 = 0.96$: sell 0.96 units of f per unit of s
- Step 3: $HE = \rho^2 = 0.64$; the hedged volatility is $\sigma_s\sqrt{1-\rho^2} = 0.72\%$ instead of 1.2%
- If tomorrow DCC forecasts $\rho_t = 0.5$ and $\sigma_{f,t} = 2\%$, the hedge ratio becomes $0.5 \cdot 1.2/2 = 0.3$: the position must be adjusted

Hedging the BET with the DAX, 2015–2026



- Daily returns on common trading days since 2005; all parameters estimated on 2005–2014; DCC: $h_t = \hat{\rho}_t \hat{\sigma}_{BET,t} / \hat{\sigma}_{DAX,t}$; rolling OLS: the slope over the previous 250 days

TSA_ch14_hedging

Interpreting the hedge ratios

- The static ratio 0.51 reflects 2005–2014, which includes the 2008 crisis; after 2015 the DCC ratio averages 0.29, from 0.07 to 1.00 (20 December 2018)
- Hedging effectiveness out of sample (2 893 days): static 13.6%, rolling OLS 18.0%, DCC 19.1%
  - the dynamic ratios remove more risk, and DCC does so without choosing a window
- All effectiveness values are modest: the BET–DAX DCC correlation averages 0.32 since 2015, so ρ^2 is small
  - a daily DCC ratio also implies daily trading costs, which are not counted here

Recap: Hedging

- $h_t^* = h_{sf,t}/h_{ff,t} = \rho_t \sigma_{s,t}/\sigma_{f,t}$; $HE = 1 - \text{Var(hedged)}/\text{Var(unhedged)}$
- BET with DAX: DCC 19.1% against static 13.6% out of sample
- Judge a hedge out of sample and after trading costs

Links with Chapters 5, 6 and 7

- **Chapter 5**
  - every MGARCH model contains univariate GARCH models
  - in DCC they are literally step 1
- **Chapter 6**
  - a VAR models spillovers in the **mean** (e.g. whether yesterday's DAX return helps to forecast today's BET return)
  - BEKK with a full **A** models spillovers in the **variance**
  - the two combine: a VAR for $\boldsymbol{\mu}_t$ and an MGARCH for $\mathbf{H}_t$; spillover indices from variance decompositions Diebold and Yilmaz (2009)
- **Chapter 7**
  - cointegration is a **long-run** link between price levels
  - correlation is a **short-run** link between returns
  - two prices can be highly correlated day by day and drift apart for ever, or cointegrated with low daily correlation
  - a VECM can carry a DCC error term: equilibrium in the mean, dynamic correlation in the shocks

Limits and extensions



Frankfurt Stock Exchange

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- One (a, b) for all pairs; no spillovers between volatilities in DCC
- Normal QML underestimates the tails: use Student t innovations or FHS for risk
- Asymmetry: correlations react more to joint falls (asymmetric DCC, Cappiello, Engle and Sheppard (2006))
- Hundreds of assets: factor models, composite likelihood and shrinkage of $\bar{\mathbf{Q}}$
- Correlation is linear dependence; for joint extremes see copulas and tail dependence Longin and Solnik (2001)

Possible contribution of AI

- **Code**
  - a first draft of the DCC likelihood, of a BEKK filter, of a Kupiec test
- **Explanation**
  - a second explanation of $\mathbf{D}_t\mathbf{R}_t\mathbf{D}_t$, of correlation targeting, of the hedge-ratio formula
- **Exploration**
  - DCC for many pairs (CEE markets, sectors, currencies) and their behaviour in crises
- Example prompt
  - `Write Python code that fits a GARCH(1,1) to daily BET and DAX log returns on common trading days, estimates a DCC(1,1) model by two-step maximum likelihood with correlation targeting, plots the dynamic hedge ratio and compares its out-of-sample hedging effectiveness with static OLS.`

Necessary checks

- The library: `arch` has no DCC model; an AI answer that calls `arch_model(..., vol='DCC')` has invented it
- The timing: $\mathbf{Q}_t$ must use $\mathbf{z}_{t-1}$, not $\mathbf{z}_t$; otherwise the VaR and the hedge use tomorrow's information
- The rescaling: $\rho_{ij,t} = q_{ij,t} / \sqrt{q_{ii,t} q_{jj,t}}$, and every $\mathbf{R}_t$ must have ones on the diagonal and $|\rho| < 1$
- The calendars: align prices on common trading days before computing returns; watch asynchronous closes
- The evaluation: estimate on one period and test on another; simulate a DCC with known (a, b) and check that the code recovers it
- Every cited reference: it must exist; check the DOI

Key takeaways

- Portfolio risk, hedging and contagion depend on time-varying covariances $\mathbf{H}_t$
- VEC is general but has too many parameters; BEKK is positive definite by construction and measures spillovers
- CCC and DCC split $\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$ and are estimated in two steps; DCC needs only (a, b) in step 2
- Real correlations move: integration, crises, new assets; constant correlation is rejected
- Applications: portfolio VaR 1% with backtesting; dynamic hedge ratios judged out of sample

Key formulas

Quantity	Formula
Correlation	$\rho_{ij,t} = h_{ij,t} / \sqrt{h_{ii,t} h_{jj,t}}$
BEKK(1,1)	$\mathbf{H}_t = \mathbf{C}\mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-\mathbf{1}} \boldsymbol{\varepsilon}_{t-\mathbf{1}}^\top \mathbf{A} + \mathbf{B}^\top \mathbf{H}_{t-\mathbf{1}} \mathbf{B}$
CCC, DCC	$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t, \quad \mathbf{D}_t = \operatorname{diag}(\sigma_{i,t})$
DCC	$\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a\, \mathbf{z}_{t-\mathbf{1}} \mathbf{z}_{t-\mathbf{1}}^\top + b\, \mathbf{Q}_{t-\mathbf{1}}, \quad \rho_{ij,t} = q_{ij,t} / \sqrt{q_{ii,t} q_{jj,t}}$
Half-life	$\ln 0.5 / \ln(a + b)$
Portfolio VaR 1%	$-(\mathbf{w}^\top \boldsymbol{\mu} + q_{\mathbf{0.01}} \sqrt{\mathbf{w}^\top \mathbf{H}_t \mathbf{w}})$
Hedge ratio	$h_t^* = h_{sf,t} / h_{ff,t}, \quad HE = 1 - \operatorname{Var}(r_s - h r_f) / \operatorname{Var}(r_s)$

Self-assessment (1/2)

- **Question**: how many parameters does a diagonal BEKK(1,1) have for $N = 3$?
  - **Answer**: $N(N+1)/2 + 2N = 6 + 6 = 12$
- **Question**: $h_{\mathbf{11}} = 1$, $h_{\mathbf{22}} = 4$, $h_{\mathbf{12}} = 1.2$. What is the correlation, and is the matrix positive definite?
  - **Answer**: $\rho = 1.2/\sqrt{4} = 0.6$; yes, $1 \cdot 4 - 1.44 > 0$
- **Question**: DCC with $a = 0.02$, $b = 0.97$, $\bar{q}_{\mathbf{12}} = 0.4$, $q_{\mathbf{11},t-\mathbf{1}} = q_{\mathbf{22},t-\mathbf{1}} = 1$, $q_{\mathbf{12},t-\mathbf{1}} = 0.4$ and $z_{t-\mathbf{1}} = (1, -1)$. What is $q_{\mathbf{12},t}$?
  - **Answer**: $0.01 \cdot 0.4 + 0.02 \cdot (-1) + 0.97 \cdot 0.4 = 0.372$; opposite moves lower the correlation
- **Question**: what is the half-life of correlation shocks when $a + b = 0.95$?
  - **Answer**: $\ln 0.5 / \ln 0.95 \approx 13.5$ days

Self-assessment (2/2)

- **Question**: $\sigma_s = 2\%$, $\sigma_f = 1\%$, $\rho = 0.5$. What are the hedge ratio and the hedging effectiveness?
  - **Answer**: $h^* = 0.5 \cdot 2/1 = 1$; $HE = 0.25$
- **Question**: a VaR 1% has 26 violations in 2 500 days, 10 of them in one month. Is it a good VaR?
  - **Answer**: the frequency is fine (expected 25), but the violations cluster: it fails the independence test and does not adapt to crises
- **Question**: why can the daily correlation of the S&P 500 and the BET be lower than the weekly one?
  - **Answer**: asynchronous trading: news from New York reaches Bucharest the next day, so daily returns of the same day do not match
- **Question**: does a higher correlation in a crisis prove contagion?
  - **Answer**: not by itself: higher volatility raises the sample correlation mechanically Forbes and Rigobon (2002); compare correlations of standardised residuals
- Next: Chapter 15, review and exam preparation; check yourself with the quiz of this chapter

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